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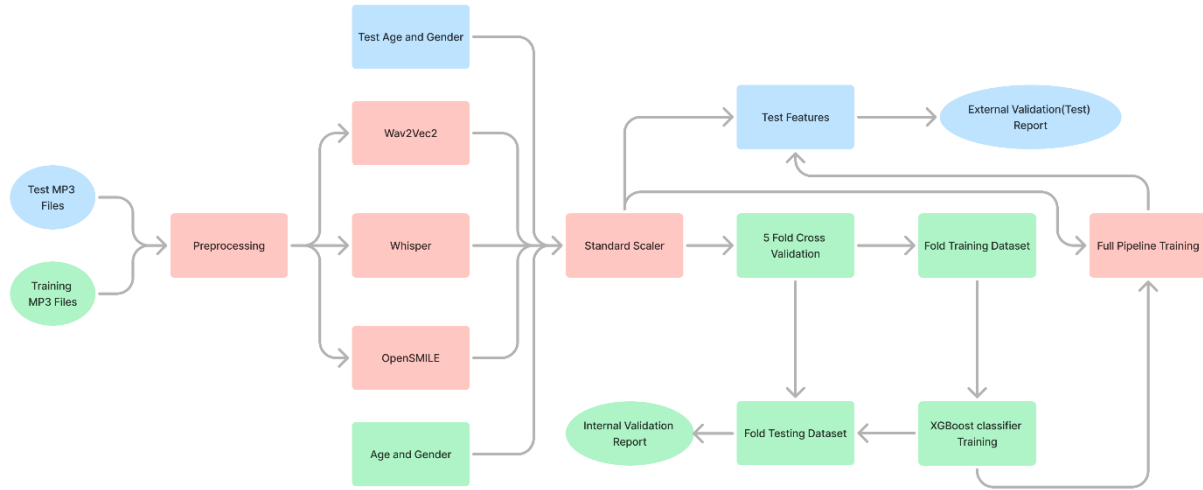
## **PREPARE Challenge Acoustic Track Solution Report**

### **Abstract**

Early and accurate diagnosis of Alzheimer's disease (AD) and Mild Cognitive Impairment (MCI) is essential for timely intervention and effective management. This report presents a machine learning framework for classifying audio recordings into three categories: Normal (NL), MCI, and AD. We developed a feature extraction pipeline that integrates traditional acoustic features from OpenSMILE's eGeMAPSv2, advanced embeddings from transformer-based models Wav2Vec2 and Whisper, and demographic information including age and gender. An optimized XGBoost classifier was employed to perform the multiclass classification task. The model's performance was evaluated using five-fold GroupKFold cross-validation on an internal dataset and subsequently tested on an external test dataset, demonstrating consistent and improved metrics such as log loss and F1-score. Class-specific analysis revealed high sensitivity in the NL class and high specificity in both MCI and AD classes, highlighting the model's strong ability to correctly identify non-impaired individuals and reliably exclude non-cases of MCI and AD. However, sensitivity in the MCI and AD classes remains an area for further enhancement. Comprehensive error analysis indicated that higher misclassification rates were associated with older age, Spanish language audio samples, and data from specific collection centers, underscoring the need for standardizing data acquisition protocols and model refinement. To facilitate practical application, we developed a user-friendly interface using Gradio, enabling seamless use of the model. This framework demonstrates significant potential for supporting the early detection and diagnosis of dementia-related disorders in clinical settings.

### **Methodology**

In this project, as demonstrated in figure 1, we implemented and validated an audio preprocessing, feature extraction, and machine learning pipeline to analyze a diverse set of MP3 files and classify them into 3 classes of normal/healthy, Mild Cognitive Impairment (MCI), and Alzheimer's Disease (AD) for PREPARE Challenge, Acoustic Track.



*Figure 1. Data pipeline regrading, preprocessing, feature extraction, training, and validation of the model. Blue color represents Tests data, green color represents training data, and red represents mutual steps.*

## Preprocessing and Feature extraction

Initially, all audio files were loaded and resampled from their original 48,000 Hz to a standardized 16,000 Hz to ensure proper size for the feature extraction models. We employed the OpenSMILE toolkit<sup>1</sup> to extract the extended Geneva Minimalistic Acoustic Parameter Set (eGeMAPSv2)<sup>2,3</sup>, encompassing features such as Mel-Frequency Cepstral Coefficients (MFCCs), chroma features, spectral centroid, spectral roll-off, zero-crossing rate, and pitch, which provide a comprehensive characterization of the audio signals. Complementing these features, we integrated high-level embeddings from state-of-the-art transformer-based models, specifically Facebook's Wav2Vec2<sup>4</sup> and OpenAI's Whisper<sup>5,6</sup>; embeddings were obtained by mean-pooling the hidden states from the last layers of each model's encoder. All extracted features as well as the available demographics data (age and gender), then normalized using StandardScaler to ensure uniform scaling across different feature dimensions. This multi-faceted approach, combining traditional acoustic measures with advanced deep learning embeddings, provides a multidimensional representation of the audio data, enhancing the potential for accurate and insightful downstream analyses.

## Feature Selection and Model Training

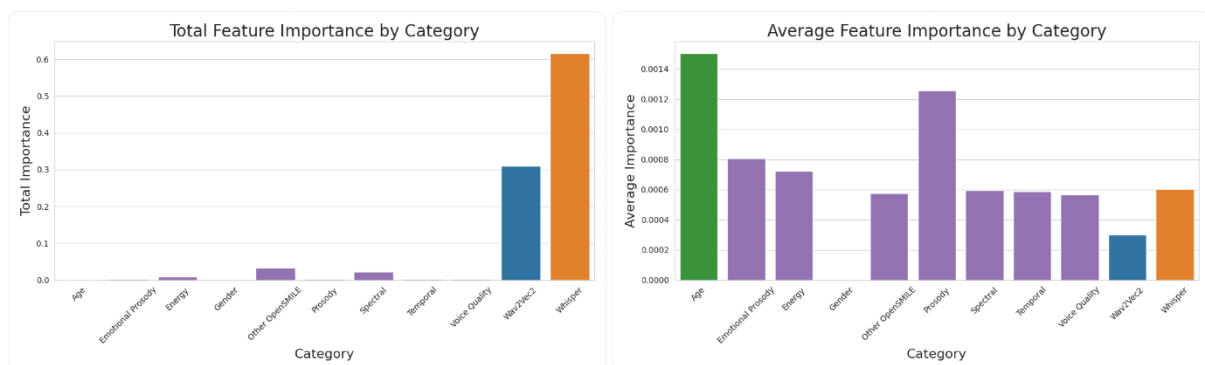
For model training and internal validation (validation on the train dataset), we utilized a structured machine learning pipeline to ensure robust and reliable performance evaluation. The target labels were encoded to transform categorical labels into numerical values suitable for classification. We applied feature selection using SciKit-learn's SelectKBest with the ANOVA F-test (with number of selected classes as hyperparameter) to retain the most significant features. We employed an XGBoost classifier<sup>7</sup> configured for multi-class classification. The model hyperparameters, including max\_depth, learning\_rate, subsample, colsample\_bytree, and reg\_lambda, were set to optimize performance while preventing overfitting.

To evaluate the model's robustness, we implemented a 5-fold GroupKFold cross-

validation strategy, ensuring that samples with the same uid were confined to a single fold to prevent data leakage. Within each fold, the pipeline was cloned and trained on the training subset, followed by predictions on the validation subset. We computed a comprehensive set of evaluation metrics, including accuracy, precision, recall, F1-score, and log loss, alongside per-class sensitivity and specificity derived from the confusion matrix. These metrics were aggregated across all folds to obtain mean and standard deviation values, providing insights into the model's stability. Finally, the pipeline was retrained on the entire dataset using the best-performing fold's configuration, ensuring that the final model benefits from all available data. This approach, combining feature scaling, selection, and advanced classification with cross-validation, underpins the reliability and efficacy of our predictive modeling framework.

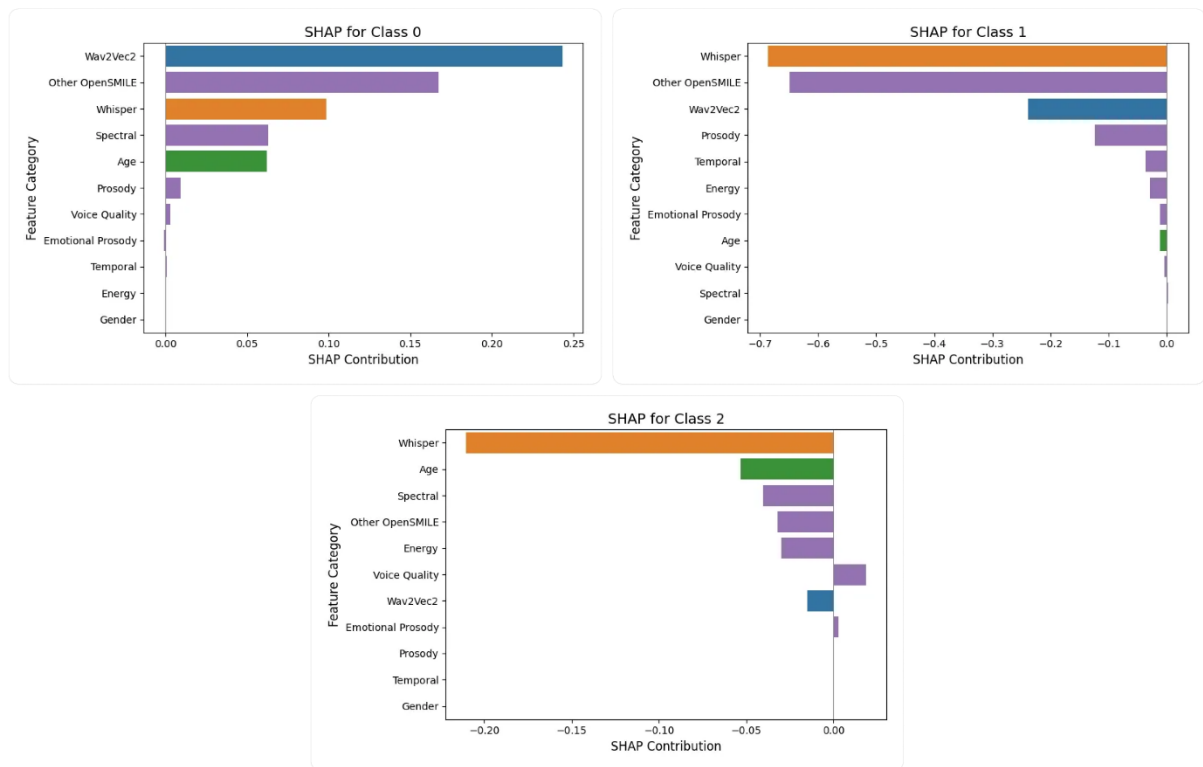
## Feature Importance and Explainability

To elucidate the contribution of different feature categories to the model's predictive performance, we conducted a detailed feature importance analysis using the trained XGBoost classifier. These features were categorized based on their origin — embeddings from Wav2Vec2 and Whisper models, OpenSMILE-derived features, and demographic variables such as age and gender.



*Figure 2. Bar plots representing categorized feature importance of all features selected for the training of the final model. Age category only represents age, other categories represent the mean of several features extracted using each feature extraction algorithms.*

To further elucidate the model's decision-making process on an individual level, we applied SHAP<sup>8</sup> to generate explanations for single subject prediction. This involved scaling and selecting the relevant features through the established pipeline, followed by calculating SHAP values for each feature across all 3 classes. The SHAP values were then aggregated by feature category, allowing us to visualize the impact of each category on the subject's classification outcome through categorized bar plots. This dual approach of global feature importance visualization and individual SHAP-based explanations provides both an overarching and granular understanding of the factors driving the model's predictions, thereby enhancing the transparency of the proposed method.



*Figure 3. SHAP analysis regarding analysis of the effect of feature categories of a single individual without Mild Cognitive Impairment or Alzheimer's Disease*

## Performance Analysis

Following comprehensive hyperparameter tuning and feature selection optimization across various models, we integrated all features derived from multiple feature extraction algorithms to construct our final feature set. Employing an XGBoost classifier configured with a maximum depth of 4 and a regularization parameter (lambda) of 20, we performed multiclass classification to differentiate between Normal (NL), Mild Cognitive Impairment (MCI), and Alzheimer's Disease (AD) classes. The model demonstrated robust and stable performance during five-fold cross-validation on the internal dataset, exhibiting consistent metrics across all folds. Upon validation with an external test dataset, the model not only maintained its performance stability but also showed slight improvements in overall and class-specific metrics, indicating good generalizability.

Moreover, Class-specific evaluations revealed that the model achieved high sensitivity in accurately identifying individuals without cognitive impairment (NL class), effectively minimizing false negatives. Additionally, the model exhibited high specificity in the MCI and AD classes, ensuring reliable exclusion of non-cases and reducing false positives. However, sensitivity in the MCI and AD classes was moderate, suggesting some challenges in correctly identifying all instances within these categories. Overall, the high specificity in MCI and AD classes coupled with high sensitivity in the NL class underscores the model's efficacy in distinguishing between different cognitive states, while also highlighting areas for future improvement to enhance sensitivity for detecting MCI and AD cases. The consistent

cross-validation results and improved external validation performance collectively affirm the model's stability and its potential for reliable application in clinical settings.

*Table 1 Performance result of the 5-fold cross validation using the training dataset demonstrated by mean and standard deviation, and the test dataset.*

| <b>Metric</b>   | Mean Internal Validation | SD Internal Validation | External Validation (Test) |
|-----------------|--------------------------|------------------------|----------------------------|
| Accuracy        | 0.68                     | 0.02                   | 0.72                       |
| Precision       | 0.68                     | 0.02                   | 0.72                       |
| Recall          | 0.68                     | 0.02                   | 0.72                       |
| F1-Score        | 0.66                     | 0.02                   | 0.7                        |
| Log Loss        | 0.72                     | 0.03                   | 0.68                       |
| NL_Sensitivity  | 0.88                     | 0.01                   | 0.91                       |
| NL_Specificity  | 0.5                      | 0.04                   | 0.53                       |
| MCI_Sensitivity | 0.31                     | 0.1                    | 0.37                       |
| MCI_Specificity | 0.97                     | 0.01                   | 0.98                       |
| AD_Sensitivity  | 0.5                      | 0.04                   | 0.52                       |
| AD_Specificity  | 0.9                      | 0.01                   | 0.92                       |

## Error and Bias Analysis

To further investigate the factors influencing the model's performance, we conducted a comprehensive error analysis on the test dataset. Our trained XGBoost model was employed to predict the class of each audio sample, and these predictions were subsequently compared to the true labels to assess overall performance and identify misclassification patterns. We segregated the audio files into two distinct groups: those correctly classified and those incorrectly classified. Statistical analyses were then performed to determine which metadata attributes differed significantly between these groups. Specifically, we utilized t-tests for numerical variables, such as age, and chi-square tests for categorical variables, including language and data collection centers.

The results, depicted in Figure 3, revealed that the incorrectly classified samples were associated with significantly higher ages ( $p\text{-value} < 0.05$ ), suggesting that age may be a contributing factor to classification errors. Additionally, audio data in Spanish exhibited a markedly higher misclassification rate (44.3%) compared to English (25%) and Mandarin (0%), with the differences being statistically significant ( $p\text{-value} < 0.05$ ). This indicates potential challenges in accurately processing non-English languages within the model. Furthermore, a significant proportion of misclassifications originated from specific data collection centers—Baycrest, Ivanova, and Delaware—highlighting possible inconsistencies or issues in data

acquisition processes at these sites. These findings underscore the importance of standardizing demographic and procedural variables to enhance the model's robustness and generalizability across diverse populations and data sources.

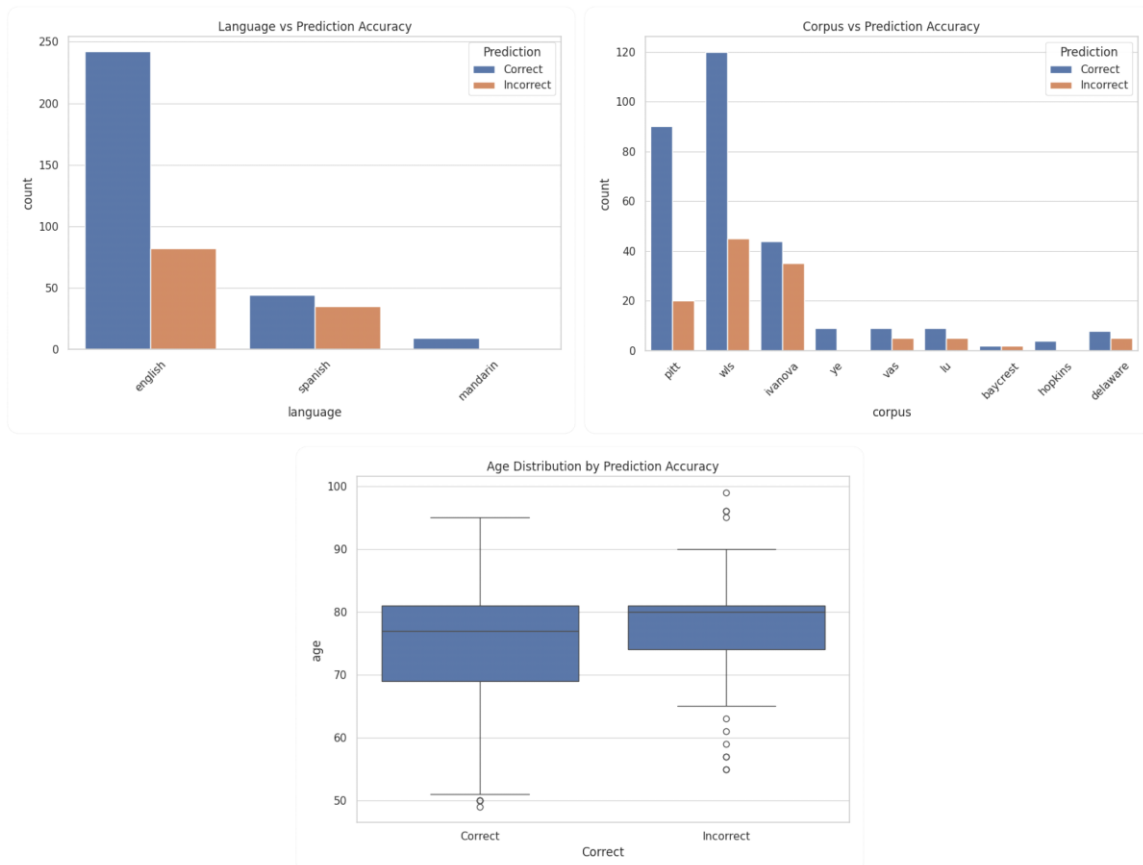


Figure 4. Demonstration of metadata that are significantly different between correct and incorrect prediction of the model in the test dataset.

## Deployment and Usability

To demonstrate the practical applicability and usability of our trained model, we developed a user-friendly interface using Gradio<sup>9</sup>, a versatile framework for creating machine learning web applications. This interface allows users to interact with the model either by recording patient voice directly through the application or by uploading a prerecorded MP3 file. Additionally, the interface requires input of demographic parameters, specifically age and gender, which are essential for accurate classification. Upon receiving the inputs, the model generates probabilistic predictions for each cognitive class (Normal, Mild Cognitive Impairment, and Alzheimer's Disease) and produces corresponding explainability charts, leveraging the SHAP-based methods detailed in this study to elucidate the model's decision-making process. The application is designed for flexible deployment, capable of running on a standard personal computer or on a remote server accessible via the internet. Importantly, the feature extraction component of the application can be deployed locally, thereby enhancing patient privacy by eliminating the need for data transmission over the internet and reducing the risk of data breaches. This ensures that sensitive patient information remains secure and confined to the local environment. The complete application, along with its feature extraction

functionalities, is accessible through the associated GitHub repository, providing an open-source platform for researchers and clinicians to evaluate and utilize the model in real-world settings. This deployment strategy not only underscores the model's feasibility for integration into clinical workflows but also facilitates broader accessibility and reproducibility, promoting its potential adoption for patient-centered diagnostic applications. All of the code, instructions, and trained model for this demo software are available in the [GitHub repository](#).

## Future Directions

Several routes remain open for enhancing the model's performance and robustness. One promising approach involves the use of ensemble methods, such as voting classifiers, which combine multiple models to achieve stronger classification performance; preliminary experiments with an ensemble voting classifier on the test dataset demonstrated a significant improvement in log loss (from 0.68 to 0.63). Further exploration of different model architectures, feature extraction techniques, and training methodologies revealed challenges, including issues with convergence and extreme overfitting, particularly when retraining pretrained deep learning models like Whisper and Wav2Vec2 on the competition's dataset.

To reduce the model's susceptibility to overfitting, various data augmentation techniques were employed, including sliding window sampling with different crop sizes of the audio files, adding background noise, and altering the tone and speed of the recordings; however, these augmentation methods did not produce significant improvements in model performance. Therefore, increasing the size and diversity of the training dataset is essential, as a larger dataset would provide the models with more varied examples, thereby enhancing their ability to generalize to unseen data. Additionally, attempts to utilize audio-to-text models for extracting text-based features through deep learning architectures were unsuccessful due to the multilingual nature of the dataset and factors such as background noise and low voice tones that impeded accurate transcription of audio to text. Future work could overcome many of these challenges by implementing more standardized data acquisition protocols, improving audio quality, and employing advanced multilingual transcription models, which would facilitate more effective feature extraction from transcribed text and potentially improve classification performance.

Overall, future research should focus on expanding and diversifying the training dataset, standardizing data acquisition processes, and refining data augmentation techniques, as these strategies hold the potential to further enhance the classification performance and generalizability of the model, making it more effective for real-world clinical applications. Additionally, as demonstrated in our [GitHub repository](#), developing an accessible software solution can increase the availability of this model to the public, thereby improving the diagnosis and potentially enabling early detection of Alzheimer's disease and other dementia-causing disorders.

## Data and Code Sharing

All of the code and the final trained model regarding the method presented in this report is publicly available in the following GitHub repository:

<https://github.com/SiavashShirzad/PREPARE-Acoustic-Track-Competition>

## References

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